

Answer Key — Exploring Forces

Final answers with brief reasons.

Objective

- Q1 **(a) newton** — force is measured in newtons (N).
-
- Q2 **(d) change of colour** — force does not change an object's colour.
-
- Q3 **(b) opposite to the motion** — friction opposes motion.
-
- Q4 **(c) gravitational** — gravity is always attractive.
-
- Q5 **(b) repel** — like charges repel; unlike attract.
-
- Q6 **(b) False** — friction needs surfaces in contact — it is a contact force.
-
- Q7 **force** — a push or pull is a force.
-
- Q8 **muscular** — muscular force comes from muscle action.
-

Short answer

- Q9 A push/pull from interaction of two objects; effects: start motion, change speed, change direction, change shape (any two).
-
- Q10 Contact forces need touching surfaces (e.g. friction); non-contact forces act at a distance (e.g. gravity).
-
- Q11 Like charges repel, unlike charges attract; the force is the electrostatic force.
-

Long / application

- Q12 Friction — contact force (needs surfaces touching). Magnetic force — non-contact (acts at a distance, can attract or repel). Gravity — non-contact and always attractive. Friction acts only on contact; gravity is always attractive.
-
- Q13 (i) Friction between the tyres/road (and air) acts opposite to motion, slowing and stopping it. (ii) Rubbing charges the balloon; the charged balloon exerts an electrostatic force that attracts it to the (uncharged) wall.
-